Sealed-bid repo on Canton

# Repo quotes, behind glass

A borrower asks a panel of lenders for cash against collateral. Each lender's rate and haircut reach the borrower and nobody else, and the repo then lives on the ledger until it is repurchased.

talang-desk.vercel.app github.com/PugarHuda/talang

Pugar Huda Mantoro

# Asking for a repo price is itself information

## 5 → 4

### The ask leaks

A fund asks five dealers for cash against a block of Treasuries. All five learn it needs liquidity and what it holds. The four that lose keep that knowledge.

## rate

### Hiding it costs

Large holders split requests across days or stay with one dealer to avoid the leak, and pay for it in rate.

## email

### After the trade, by hand

Margin calls argued from each side's own spreadsheet and price, substitutions by email, the regulator's report rebuilt after the fact.

Source: BRIEF.md. No user interviews are claimed; the validation log there records only conversations that happened, and is empty.

# A quote signed by one lender and the borrower, and nobody else

On Canton a contract is sent only to its stakeholders' nodes. A sealed quote has two, so it never reaches a rival lender or the regulator. Not hidden by the page: never received.

The quote the borrower takes opens a repo that the contract runs until it is repurchased.

| PAIN | WHAT THE LEDGER ENFORCES |
| --- | --- |
| **Quotes that cannot be funded** | Full principal locked behind the quote |
| **Settlement risk at open** | Collateral pledged, cash delivered, losers refunded: one transaction |
| **Disputed margin calls** | Units due from a signed mark; refused while covered |
| **Substitution by email** | Lender consent, and only if the substitute covers at a fresh mark |
| **Rolling = close + reopen** | Interest so far paid, new rate from today |
| **Default** | After an unanswered call or maturity, pledge to the lender |

# Sealed quotes, atomic award, proof without leaks

01

## Request

Borrower invites a panel. Only invited lenders see it, and it carries no rate.

02

## Sealed quote

Lender locks the full principal: desk escrow or a CIP-0056 allocation. Rivals and the regulator never receive it.

03

## Award

Opening leg atomic: collateral pledged, cash delivered, losing quotes refunded unrevealed.

04

## Loss notice · best execution

Each loser gets its rank and nothing else. The regulator gets quotes weighed, winner's rank, gap to best as a band (0, 1-5, 6-25, >25 bp). No lender named.




The demo case on the seeded ledger: the cheapest rate's haircut does not cover, so the borrower takes rank 2. Regulator: **3 quotes weighed, winner rank 2, gap to best in the 6-25 bp band**. The losing lender: **rank 3 of 3**. Neither learns a losing rate or a name.

# Privacy is read from each node

| NODE | RECEIVES | NEVER RECEIVES |
| --- | --- | --- |
| **Borrower** | Its requests; every quote made to it, ranked with coverage | — |
| **Lender A / B / C** | Requests it was invited to; its own quote; its own loss notice (rank only) | Rival quotes: **0** on any lender node · other lenders' loss notices: **0** |
| **Regulator** | `RepoReport` for every lifecycle event, `BestExecution` records | Requests **0** · quotes **0** · loss notices **0** |

Checked on every run: `privacy_check` from each MCP desk in `npm run e2e:mcp`, and in page and API by `npm run qa:desk`. Token rail on a participant: lender B's node held 0 rival quotes.

# No single key moves a margin call

Marks come from a valuation committee, a decentralized party. Each mark is a `GovernableAction` on BitSafe's own `GovernanceRules`, DARs used unmodified. A lender can be a k-of-n syndicate the same way.

- 1 of 3 confirmations: **refused** by BitSafe's rule *"Enough confirmations to execute action"*
- 2 of 3: mark published, repo opens on it
- Margin call on one pricer's own mark: **refused** ("mark from the wrong agent")
- Oracle: one pricer per exchange; a proposal 8% off market refused

### DecMan LocalNet · 3 Canton 3.5.8 participants

Committee onboarded through DecMan, hosted on all three nodes, threshold 2.

**PUBLISHES**

Participant 3 stopped: members on nodes 1 and 2 publish UST10Y 97,500.

**CANNOT**

Participants 2 and 3 stopped: publish stops at the first confirmation.

```
update 12203359acdb…a4138 · one node down
update 1220db9dfd14…c42b · 2-of-3 mark
```

One command: `bash localnet/demo.sh reset`. In DecMan's own UI: `media/decman-demo-captioned.mp4`. On the NODERS DevNet participant: **RAN** 1 of 3 refused, 2 of 3 executed, repo and margin call on the committee mark, 8 / 8 steps, 9 October (all parties on one node: shared control, not distributed hosting).

# Real CBTC as collateral, through the DA Utility registry

| STEP ON THE NODERS DEVNET PARTICIPANT | UPDATE ID |
|---|---|
| Faucet's 1.0 CBTC transfer offer accepted through the registry | `12085841bcd…f6c5` |
| CBTC allocated to lender A via the registry's `AllocationFactory` | `1220c89b7d65…d45c` |
| Request: 50 USDC against 0.001 CBTC, 1 day | `1220f040d4b6…ee2e` |
| Sealed quote, principal escrowed | `12205686d3f2…49f9` |
| Award: USDC to borrower, CBTC allocation held | `1220dde0a7ac…8746` |
| Repurchase: allocation cancelled via the registry | `1220db50b682…24dd` |

Choice context and disclosed contracts come from the registry's off-ledger API (`lib/registry.mjs`). Borrower holds 1 CBTC again after repurchase. 7 of 7 steps, 9 October.

Cash leg here is desk USDC. **Both legs in real tokens**, same participant: Canton Coin cash (DevNet tap, quote funded by an Amulet allocation through the Scan registry) and CBTC collateral; award, repurchase, CBTC home. 12 of 12 steps, `token-repo-amulet-cbtc-devnet.json`.

**Default in real CBTC**: margin call left unanswered, lender A defaults, 0.001 CBTC executed to it through the registry, offset 3209860 (9 and 12 October, `cbtc-default-devnet.json`).

USDCx on DevNet: `NOT YET` no DevNet USDCx registry or faucet documented.

# The wallet's key authorises the desk

Grofty signs only standard token operations on its own participant, so Talang does not route Daml commands through it. A desk role bound to a wallet acts only after the wallet signs a single-use challenge naming the exact command (CIP-0103 `signMessage`).

01

### Challenge

A single-use nonce that names the exact desk command the role is about to submit.

02

### Verify

`api/wallet-verify.mjs` checks nonce, signer, Ed25519 / P-256 signature, and that the key's fingerprint is the party's namespace. Then submits.

03

### Receipt

The borrower can verify which wallet key sealed a lender's quote; rival lenders see nothing. Venue fee payable in USDCx or CC on TestNet / MainNet.

**RAN** With the live Grofty extension on TestNet, 12 October: connect; two borrower commands signed in Grofty and verified (executed on a local sandbox: Grofty's participant cannot host Talang's package); 1 CC venue fee executed on TestNet, update `1220e6e3…ccad`, to Grofty's QA party as a stand-in venue.

# An agent at each desk, bound by the same privacy

An MCP agent reads only its party's node. A lender's agent never sees a rival's rate; the regulator's sees reports and nothing upstream. Every action is re-checked by the contract.

**lenderA / B / C**

`portfolio · open_requests · quote · call_margin · review_substitution · offer_roll · loss_history · privacy_check`

**borrower**

`quotes (ranked, with coverage) · award ("best" = cheapest covered) · book · accept_roll · privacy_check`

**regulator**

`lifecycle · best_execution · privacy_check`

`npm run e2e:mcp`: 41 of 41 checks on a Canton participant, every desk end to end, privacy read from each node. Also in CI.

# A venue fee that cannot be skipped

## 10 bp

per annum of principal, ACT/360, collected inside the repurchase and on every roll.

9.5M repo held 60 days → **1,583.33** to the venue

- **Desk cash:** a split of the borrower's repurchase payment.
- **CIP-0056 asset:** a second leg of the repurchase `AllocationRequest`, under the operator's standing `VenueAgreement`. On a participant: 50,008.33 USDCx to the lender, 0.14 to the venue.
- **Who pays:** borrowers (funds, treasuries holding tokenized bonds, T-bills or CBTC) who want cash without advertising it.
- **Status:** `BUILT` tested in `venueFeeOnRepurchase`, `tokenLifecycle`. No revenue yet.

# What ran, and where

**31 / 31**
Daml scripts: lifecycle, refusals, tokens, BitSafe, 1.2.0

**41 / 41**
MCP end-to-end checks, local sandbox and CI

**29 / 29**
Playwright desk QA, every role and button

**45 / 45**
Read-only proxy checks: no secret, no submit

| RUN | WHERE | RESULT |
|---|---|---|
| **`talang-repo` 1.0.0 seeded; 1.1.0, 1.2.0 vetted** | NODERS DevNet `hackcanton-devnet-3` | Hosted desk reads it live |
| **CBTC rail · CBTC default** | NODERS DevNet, DA Utility registry | 7 / 7 · 9 / 9 steps, update ids recorded |
| **Canton Coin cash + CBTC collateral** | NODERS DevNet, Scan + DA Utility registries | 12 / 12 steps, update ids recorded |
| **BitSafe 2-of-3 marks · node offline** | DecMan LocalNet, 3 participants; marks also on DevNet | 8 / 8 · 6 / 6 steps; DevNet 8 / 8 |
| **CIP-0056 token rail · price oracle** | Local Canton 3.4.11 participant | 8 / 8 · 4 / 4 steps |
| **Grofty wallet, live extension** | Canton TestNet (fee) + local sandbox (commands) | 2 commands signed, 1 CC fee executed |

CI: GitHub Actions run 38272515745, commit f9e539b. Evidence JSON in `docs/evidence/`. Users or pilots: none yet.

# The product is the privacy model

| | PUBLIC EVM CHAIN | PRIVACY L2 / OBJECT CHAIN | CANTON |
|---|---|---|---|
| **Who sees a lender's rate** | Everyone: calldata and mempool | Hidden behind ZK, encryption or an enclave the desk builds and audits | **The borrower who asked.** No other stakeholder, no other node receives it |
| **Opening leg** | Atomic, both amounts public | Atomic on its chain; other issuers' assets via a bridge | **One Daml transaction** across both legs, incl. CIP-0056 assets from separate registries |
| **Regulator view** | Same public data, or an off-chain report | Viewing keys or a separate channel | **Observer** on `RepoReport` and `BestExecution`; absent from quotes |
| **k-of-n governance** | Multisig; votes and prices public | Multisig or committee per chain | **BitSafe `GovernanceRules`** on a party hosted on several nodes; survives one of three down |

# From a shadow book to live repos

2 weeks

## Shadow book on DevNet

One borrower, two lenders run their real collateral list and rates without moving value. Measure request-to-award time, quotes per request, calls the contract would issue.

4 weeks

## Small live repos on MainNet

Collateral from one tokenized-bond or CBTC registry, cash in USDCx, both through CIP-0056. Ticket size capped.

then

## Independent marks, regulator node

Committee run by independent price providers as a BitSafe decentralized party; an auditor runs an observer node.

NOT YET USDCx on DevNet · a Talang command executed on TestNet · any pilot user. Pilot success: award in under 10 minutes, two or more lenders quoting, zero rival quotes on any lender node.

# What would move this to a pilot


## Pugar Huda Mantoro

Builder. Prior work: Tirai, a confidential multi-dealer RFQ desk on Canton (2nd place, HackCanton Season 2). Talang reuses its escrow-on-quote and CIP-0056 settlement patterns; first Talang commit 2 October 2026.

`github.com/PugarHuda`


**NODERS** intros to two tenants for a shadow book

**Pilot** one borrower and two lenders for a two-week shadow book

**Registries** a USDCx registry on DevNet, a tokenized-bond or T-bill registry

**Prices · wallet** a price provider to sit on the committee; a TestNet participant that hosts Talang's package